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## Authentication of hardware component firmware


### Abstract

An example apparatus may include a hardware component, a memory, and a processor. The processor may retrieve a firmware for the hardware component subsequent to the firmware being loaded on the hardware component. The processor may determine that a firmware signature associated with the firmware is stored in the memory. In some examples, in response to a determination that the firmware signature associated with the firmware is stored in the memory, the processor may initiate authentication of the firmware.



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## Background/Summary

BACKGROUND

(1) Electronic devices, including computing devices, may be made up of many different hardware components. These hardware components may include firmware to support various features on the electronic devices, and in some cases may support firmware updates.

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## Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Features of the present disclosure are illustrated by way of example and not limited in the following figure(s), in which like numerals indicate like elements, in which:

(2) FIG. **1** depicts a block diagram of an example apparatus that may initiate authentication of firmware for a hardware component;

(3) FIG. **2** shows a block diagram of an example apparatus that may initiate a recovery process to recover firmware for a hardware component;

(4) FIG. **3** shows a flow diagram of an example method for authenticating firmware and allowing/disallowing startup/connection of a hardware component; and

(5) FIG. **4** depicts a block diagram of an example non-transitory computer-readable medium that may have stored thereon computer-readable instructions to initiate authentication of firmware for a hardware component.

DETAILED DESCRIPTION

(6) Disclosed herein are apparatuses, systems, methods, and computer-readable media that may initiate authentication of a hardware component firmware (hereinafter simply referred to as firmware). Electronic devices, such as personal computers, may be made up of many different hardware components, including cameras, a mouse or mouse-like glide and click pads, hardware interfaces/connectors, or the like. Many of these hardware components may have firmware and may support firmware updates that may correct bugs, add features, address security issues, change hardware component behavior, and/or the like.

(7) As used herein, a firmware (or a device firmware) refers to instructions to provide low-level control for a hardware component's specific hardware. The firmware may provide a standardized operating environment for more complex device software (e.g., an operating system of a computing device), or may operate as an operating system for the hardware component, performing control, monitoring, data manipulation functions, and/or the like. The firmware may be stored in a non-volatile memory of the hardware component such as read-only memory (ROM), flash memory, and/or the like.

(8) In some examples, a firmware for a hardware component may be authenticated and/or recovered to an authenticated version of the firmware, for instance, in cases where a firmware update mechanism is not supported by the firmware, where a hardware component supports a firmware update mechanism(s) that does not support authentication, or the like. In this regard, a processor may authenticate the firmware in cases where that firmware may have been loaded (e.g., installed or updated) in a less secure context, for instance, in an OS context that may be more vulnerable to attacks. In some examples, a Basic Input/Output System (BIOS) may be implemented to allow or deny initialization, control, and operation of the hardware component based on a signature verification of the firmware.

(9) As used herein, a BIOS refers to hardware or hardware and instructions to initialize, control, or operate a computing device prior to execution of an OS. Instructions included within a BIOS may be software, firmware, microcode, or other programming that defines or controls functionality or operation of a BIOS. In one example, a BIOS may be implemented using instructions, such as platform firmware of a computing device, executable by a processor. A BIOS may operate or execute prior to the execution of the OS of a computing device. A BIOS may initialize, control, or

operate components such as hardware components of a computing device and may load or bootup the OS of computing device.
(10) In some examples, a BIOS may provide or establish an interface between hardware devices or platform firmware of the computing device and an OS of the computing device, via which the OS of the computing device may control or operate hardware devices or platform firmware of the computing device. In some examples, a BIOS may implement the Unified Extensible Firmware Interface (UEFI) specification or another specification or standard for initializing, controlling, or operating a computing device.
(11) During bootup, the BIOS may first perform various pre-OS operations including, for instance, a Power-On Self-Test (POST), and may then proceed to load the OS. During the bootup process, the BIOS or an embedded controller (EC) may retrieve a firmware for a hardware component connected to an electronic device, such as a computer. The BIOS or the embedded controller may determine whether a firmware signature associated with the retrieved firmware is stored in a memory, for instance, in a private Serial Peripheral Interface (SPI) of the embedded controller. In response to a determination that the firmware signature associated with the retrieved firmware is stored in the memory, the BIOS or the embedded controller may initiate authentication of the firmware.
(12) In some examples, the BIOS (or the embedded controller) may determine that the firmware is inauthentic in response to the firmware signature associated with the firmware being absent from the memory of the embedded controller. In this instance, the BIOS may initiate a recovery process to recover the firmware for the hardware component. In some examples, based on a determination that a firmware update for the hardware component is stored in the memory, the processor may execute the BIOS to replace the firmware with the firmware update at the hardware component and authorize the hardware component to establish connection to a bus based on the firmware update. In some examples, the BIOS may disable the hardware component or prevent connection of the hardware component, for instance to a bus (e.g., a PCI bus, USB bus, or the like), and/or may generate a notification recommending recovery of the firmware when authentication fails.
(13) A concern associated with some hardware components may be that a firmware update mechanism(s) supported by the hardware components may not support any mechanism for authenticating the firmware updates or may support both an authentication mechanism and insecure updates (e.g., legacy update mechanisms that do not support authentication), and in such instances, some hardware component firmware updates may be insecure. By enabling, for instance, a BIOS/EC to authenticate a device firmware, the BIOS/EC may be implemented to disallow a device firmware loaded using unsupported firmware update mechanisms. In this regard, the BIOS/EC may invalidate insecure updates, for instance, via recovery, device disablement, notifications, and/or the like, and thereby provide a mechanism to recover to a firmware that has been authenticated. The apparatuses, systems, methods, and computer-readable media discussed herein may improve security for hardware components on legacy firmware update mechanisms or unsupported firmware update mechanisms, for instance, when the hardware component itself supports no authentication mechanism or when the hardware component presents a firmware update mechanism in a less secure context.
(14) By allowing the BIOS/EC to authenticate the device firmware at bootup, security of the system, including security for legacy devices that may not support a mechanism for firmware updates, may be improved by automatically disabling a hardware component once an attack is discovered, or in some cases, automatically recovering the firmware to an authenticated firmware version. As such, firmware recovery and update may be implemented in a relatively quick and efficient manner, which may enable a computing device on which the BIOS/EC is executed to operate in an efficient and secure manner.
(15) Reference is first made to FIG. **1**, which shows a block diagram of an example apparatus **100** that may initiate authentication of firmware. It should be understood that the apparatus **100**

depicted in FIG. **1** may include additional features and that some of the features described herein may be removed and/or modified without departing from the scope of the apparatus **100**.

(16) The apparatus **100** may include a processor **102**, a hardware component **104**, and a memory **106**. The apparatus **100** may be a server, a node in a network (such as a data center), a personal computer, a laptop computer, a tablet computer, a smartphone, a network gateway, a network router, an electronic device such as Internet of Things (IoT) device, and/or the like. The processor **102** may be a semiconductor-based microprocessor, a central processing unit (CPU), an application-specific integrated circuit (ASIC), a field-programmable gate array (FPGA), and/or other hardware device. Although the apparatus **100** is depicted as having a single processor **102**, it should be understood that the apparatus **100** may include additional processors and/or cores without departing from a scope of the apparatus **100**. In this regard, references to a single processor **102** as well as to a single memory **106** may be understood to additionally or alternatively pertain to multiple processors **102** and multiple memories **106**.

(17) The hardware component **104** may be, for example, a device loaded on the apparatus **100** and may include a camera, a keyboard, a computer mouse, a mouse-like glide and click pad, a data connector, a disk drive, a video controller, or the like. In some examples, the hardware component **104** may be integrated in a single housing with the processor **102** and memory **106**, or in some examples, the hardware component **104** may be connected externally to the housing, for instance, using an expansion port, or the like.

(18) The hardware component **104** may include firmware loaded onto the hardware component **104**. The firmware may enable various features and functions on the hardware component **104**. In this regard, the firmware may generally be defined as a set of instructions that may provide low-level control for the hardware component in and/or attached to the apparatus **100**. In some instances, the firmware may provide a standardized operating environment for more complicated software, such as an operating system, in the apparatus **100**. In some examples, the hardware component **104** may support firmware updates that may correct device bugs, add device features, fix security problems, enable various changes to device behaviors, or the like.

(19) The memory **106** may be, for example, a non-volatile memory such as, Read-Only Memory (ROM), flash memory, solid state drive, Random-Access memory (RAM), an Electrically Erasable Programmable Read-Only Memory (EEPROM), a storage device, an optical disc, or the like. By way of example, the memory **106** may be non-volatile random-access memory (NVRAM), which may be implemented to store and return data over a serial programmable interface (SPI) bus/connection. The memory **106** may be implemented on an embedded controller, for instance, in the form of an SPI chip. In some examples, the memory **106** may store information to authenticate and support update of the firmware for the hardware component **104**. For instance, the memory **106** may store a firmware signature associated with the firmware.

(20) As shown in FIG. **1**, the processor **102** may perform various operations **110-114** to initiate authentication of firmware of the hardware component **104**. The operations **110-114** may be hardware logic blocks that the processor **102** may execute. In other examples, the operations **110-114** may be computer-readable instructions, e.g., non-transitory computer-readable instructions. In other examples, the apparatus **100** may include a combination of instructions and hardware logic blocks to implement or execute functions corresponding to the operations **110-114**.

(21) The processor **102** may execute the operation **110** to retrieve a firmware for the hardware component **104** subsequent to the firmware being loaded on the hardware component **104**. In this regard, the firmware may be installed during manufacture, updated using update mechanisms including unsupported or insecure update mechanisms (e.g., update mechanisms that do not support authentication of the firmware), or the like, to load an image of the firmware on the hardware component **104**.

(22) The processor **102** may execute the operation **112** to determine that a firmware signature associated with the firmware is stored in the memory **106**. In some examples, the memory **106** may

be a memory accessible during bootup. In response to a determination that the firmware signature associated with the firmware is stored in the memory **106**, the processor **102** may execute the operation **114** to initiate authentication of the firmware. In this regard, the processor **102** may verify the firmware with the firmware signature. In response to verification of the firmware, the processor **102** may allow or disallow the hardware component **104** to start up and/or to connect to a bus of the computing device, such as a PCI bus, USB bus, or the like. In some examples, the processor **102** may determine that the firmware is not authentic in response to a determination that the firmware signature associated with the firmware is not stored in the memory **106**. In this regard, based on a determination that a firmware signature is not stored in the memory **106**, the processor **102** may disallow the hardware component to start up, shut down the hardware component, recover to a firmware that has been authenticated, generate a notification to an end user or to a system administrator via the OS, and/or the like.

(23) Reference is now made to FIG. **2**, which shows a block diagram of an example apparatus **200** that may initiate a recovery process to recover firmware **212** for a hardware component **210**. It should be understood that the apparatus **200** depicted in FIG. **2** may include additional features and that some of the features described herein may be removed and/or modified without departing from the scope of the apparatus **200**.

(24) The apparatus **200** may include a processor **202**, a BIOS **204**, an embedded controller **206** having a memory **208**, and a hardware component **210**. The processor **202** may be equivalent to the processor **102**, the memory **208** may be equivalent to the memory **106**, and the hardware component **210** may be equivalent to the hardware component **104** depicted in FIG. **1** and may thus respectively include the components described in FIG. **1** with respect to the processor **102**, memory **106**, and the hardware component **104**.

(25) According to examples, the processor **202** may execute the BIOS **204** to initialize, control, and/or operate the apparatus **200** and the hardware component **210**. During apparatus **200** bootup, the processor **202** may cause the BIOS **204** to first perform a Power-On Self-Test (POST) and may then proceed to load the OS. In some examples, the processor **202** may cause the BIOS **204** to authenticate the firmware **212** during POST. In some examples, the processor **202** may cause the BIOS **204** to initiate authentication of the firmware **212** at every bootup, periodically (e.g., at every cold bootup), on-demand, or the like.

(26) The processor **202** may execute the BIOS **204** to retrieve the firmware **212** from the hardware component **210** during bootup. The firmware **212** may have been previously loaded onto the hardware component **210**, for instance, during manufacture or during a previous firmware update process.

(27) The processor **202** may determine whether a firmware signature **214** associated with the firmware **212** is stored in the memory **208**. In some examples, the firmware signature **214** may be stored on the memory **208** through a trusted firmware update mechanism. As such, in examples in which unsupported or insecure firmware update mechanisms are used to update the firmware **212**, the firmware signature **214** that corresponds to the firmware **212** may be absent from the memory **208**. In this regard, the processor **202** may detect insecure firmware updates during a subsequent bootup, based on an absence of the corresponding firmware signature **214**.

(28) In response to a determination that the firmware **212** is inauthentic, the processor **202** may initiate a recovery process to recover the firmware **212** for the hardware component **210**. In some examples, the recovery process may include the processor **202** causing the BIOS **204** to disable and/or disconnect the hardware component **210**, automatically updating the firmware **212** to a trusted version, or generating a notification indicating a status of the firmware authentication. As such, an attack initiated through, for instance, an unsupported firmware update mechanism cannot be persisted as the processor **202** may disable or disconnect the hardware component **210** upon detection of the attack, restore the firmware **212** to a firmware version, and/or generate a notification indicating a status of the firmware to the end user and/or a system administrator via an

BIOS/EC-to-OS alerting interface (e.g., a firmware audit log).

(29) In response to a determination that the firmware signature **214** associated with the firmware **212** is stored in the memory **208**, the processor **202** may initiate authentication of the firmware **212**. In this regard, the processor **202** may retrieve the firmware signature **214** from the memory **208** of the embedded controller **206** to the BIOS **204** to authenticate the firmware **212**.

(30) In some examples, the processor **202** may cause the BIOS **204** to determine that the firmware is authentic or inauthentic based on a verification process to authenticate the firmware **212** using the firmware signature **214**. In this regard, the verification process may include use of hash functions, a private key-pair, or the like, for authentication of the firmware **212**. Once the authenticity of the firmware **212** has been verified, the processor **202** may execute the BIOS **204** to allow the hardware component **210** to startup during bootup of the apparatus **100**. In some examples, the BIOS **204** may allow the hardware component **210** to connect to a bus **216** of the computing device.

(31) In some examples, the processor **202** may determine that the firmware **212** is not authentic in response to a determination that the firmware signature **214** associated with the firmware **212** is not stored in the memory **208**. In this regard, an absence of the firmware signature **214** may be an indication that the firmware **212** is not secure/trusted.

(32) In response to a determination that the firmware **212** is inauthentic, the processor **202** may cause the BIOS **204** to initiate a recovery process to recover the firmware **212** for the hardware component **210**. In some examples, to initiate the recovery process, the processor **202** may cause the BIOS **204** to disallow connection of the hardware component **210** to the bus **216**, disable the hardware component **210**, replace the firmware **212** for the hardware component **210**, and/or generate a notification indicating an authentication status of the firmware **212** of the hardware component **210**.

(33) In this regard, the BIOS **204** may perform various combinations of operations to disable a hardware component **210** associated with an unauthorized firmware **212**. For instance, the BIOS **204** may shut down the hardware component **210** when the firmware **212** cannot be verified using the firmware signature **214** or prevent the hardware component **210** from starting up. In some instances, the BIOS **204** may allow the hardware component **210** to startup, but may prevent the hardware component **210** from connecting to the bus **216**. In some instances in which a firmware update **218** associated with the firmware **212** is available in the memory **208**, the processor **202** may replace the firmware **212** with the firmware update **218**. Alternatively or additionally, the BIOS **204** may allow the unauthorized firmware **212** to operate normally, while generating a notification to the user to indicate the unauthenticated status of the firmware **212**.

(34) In some examples, the processor **202** may generate the notification to include an indication that the firmware **212** is not verified or is inauthentic, recommend updating the firmware **212** to a trusted version, or the like. In some examples, the processor **202** may report that the hardware component **210** is not operational to the OS of the apparatus **200** or in a firmware audit log, for instance, to allow users to verify the situation and take appropriate action, such as, loading an authenticated firmware on the hardware component **210**. In this regard, installation of an authenticated firmware may include storing a corresponding firmware signature **214** in the memory **208**, and as such, authenticity of the firmware **212** may be verified during a subsequent bootup.

(35) In some examples, in response to a determination that the firmware **212** is not authentic, the processor **202** may initiate a firmware replacement process to automatically replace the firmware **212**. In this regard, the memory **208** may store a firmware update **218** associated with the firmware **212**. The processor **202** may execute the BIOS **204** to retrieve the firmware update **218** from the memory **208** of the embedded controller **206**, and may replace the firmware **212** with a firmware, such as the firmware update **218**. As such, the BIOS **204** may automatically recover to an authenticated version of the firmware **212** using the firmware update **218** if a firmware verification should ever fail.

(36) While the processor **202** has been described herein as causing the BIOS **204** to initiate authentication of the firmware **212** and initiate a recovery process to recover the firmware **212**, in some examples, the processor **202** may cause the embedded controller **206** to perform these operations. For instance, the processor **202** may cause the embedded controller **206** to retrieve the firmware **212** from the hardware component **210**, determine that the firmware signature **214** is stored in the memory **208**, initiate authentication of the firmware **212** using the firmware signature **214**, and/or initiate a recovery process to recover the firmware **212** using the firmware update **218**. The embedded controller **206** may recover the firmware **212** by, for instance, automatically replacing the firmware **212** with a version of the firmware **212** that is secure/trusted (e.g., has been authenticated using a firmware signature), generating notifications indicating an authentication status of the firmware **212**, or the like. In some examples, the BIOS **204** may store the firmware signature **214** and/or the firmware update **218** in a memory of the BIOS **204**, for instance, in an NVRAM, which may be implemented to store and return data over an SPI bus/connection. In some instances, the BIOS **204** may store the firmware signature **214** using a UEFI specification.
(37) Various manners in which the processor **102**, **202** may operate are discussed in greater detail with respect to the method **300** depicted in FIG. **3**. FIG. **3** depicts a flow diagram of an example method **300** for authenticating firmware **212** and allowing/disallowing startup/connection of a hardware component **104**, **210**. It should be understood that the method **300** depicted in FIG. **3** may include additional operations and that some of the operations described therein may be removed and/or modified without departing from the scope of the method **300**. The description of the method **300** is made with reference to the features depicted in FIGS. **1** and **2** for purposes of illustration.
(38) At block **302**, the processor **102**, **202** may retrieve firmware **212** for a hardware component **104**, **210**. In some examples, the firmware **212** may be stored on the hardware component **210** as a firmware image. At block **304**, the processor **102**, **202** may determine that a firmware signature **214** associated with the firmware **212** is stored in the memory **208** of an embedded controller **206**. In some examples, the firmware signature **214** may be stored in a memory of the BIOS **204**, such as an NVRAM.
(39) In response to a determination that the firmware signature **214** is stored in the memory **208**, at block **306**, the processor **202** may initiate authentication of the firmware **212**. In some examples, the processor **202** may execute the BIOS **204** to perform an authentication operation to verify the firmware **212** using the firmware signature **214**. At block **308**, based on a determination at block **306** that the firmware **212** is authentic, the processor **202** may execute the BIOS **204** to allow the hardware component **210** to startup and/or connect to the bus **216**.
(40) The processor **202** may determine that the firmware **212** is inauthentic in response to a determination that the firmware signature **214** associated with the firmware **212** is absent, at block **304**, or based on a verification process to authenticate the firmware **212** using the firmware signature **214**, at block **306**. In response to a determination that the firmware signature **214** is absent at block **304** or that the firmware **212** is unauthentic at block **306**, at block **310**, the processor **202** may determine whether a firmware update **218** associated with the hardware component **210** is stored in the memory **208** of the embedded controller **206**. In response to a determination at block **310** that the firmware update **218** associated with the firmware **212** is stored in the memory **208** of the embedded controller **206**, at block **312**, the processor **202** may cause the BIOS **204** to automatically replace the firmware **212** with the firmware update **218**. In addition, following replacement of the firmware **212** with the firmware update **218**, at block **308**, the processor **202** may execute the BIOS **204** to allow the hardware component **210** to startup and/or connect to the bus **216**.
(41) However, in response to a determination at block **310** that the firmware update **218** associated with the firmware **212** is not stored in the memory **208** of the embedded controller **206**, at block **314**, the processor **202** may cause the BIOS **204** to disallow startup and/or connection of the

hardware component **210**. In this regard, the BIOS **204** may block the startup of the hardware component **210** or shut down the hardware component **210**. In some examples, the processor **202** may cause the BIOS **204** to prevent connection of the hardware component **210** to the bus **216**. Alternatively or additionally, the processor **202** may generate a notification regarding the authentication status of the firmware **212**. Such notifications may allow an end user or a system administrator to perform a firmware update using an update mechanism that supports authentication of the firmware **212**.

(42) Some or all of the operations set forth in the method **300** may be included as utilities, programs, or subprograms, in any desired computer accessible medium. In addition, the method **300** may be embodied by computer programs, which may exist in a variety of forms both active and inactive. For example, they may exist as computer-readable instructions, including source code, object code, executable code or other formats. Any of the above may be embodied on a non-transitory computer-readable storage medium.

(43) Examples of non-transitory computer-readable storage media include computer system RAM, ROM, EPROM, EEPROM, and magnetic or optical disks or tapes. It is therefore to be understood that any electronic device capable of executing the above-described functions may perform those functions enumerated above.

(44) Turning now to FIG. **4**, there is shown a block diagram of a non-transitory computer-readable medium **400** that may have stored thereon computer-readable instructions to initiate authentication of firmware for a hardware component. It should be understood that the computer-readable medium **400** depicted in FIG. **4** may include additional instructions and that some of the instructions described herein may be removed and/or modified without departing from the scope of the computer-readable medium **400** disclosed herein. The computer-readable medium **400** may be a non-transitory computer-readable medium. The term “non-transitory” does not encompass transitory propagating signals.

(45) The computer-readable medium **400** may have stored thereon computer-readable instructions **402-406** that a processor, such as the processor **102**, **202** depicted in FIGS. **1-2**, may execute. The computer-readable medium **400** may be an electronic, magnetic, optical, or other physical storage device that contains or stores executable instructions. The computer-readable medium **400** may be, for example, Random-Access memory (RAM), an Electrically Erasable Programmable Read-Only Memory (EEPROM), a storage device, an optical disc, or the like.

(46) The processor may fetch, decode, and execute the instructions **402** to retrieve a firmware **212** of a hardware component **210** of a computing device at bootup of the computing device. In this regard, the firmware **212** may have been previously updated at the hardware component **210**. The processor may fetch, decode, and execute the instructions **404** to initiate authentication of the firmware **212** using a firmware signature **214** stored in a memory **208**. In response to a determination that an authentication of the firmware **212** has failed, the processor may fetch, decode, and execute the instructions **406** to recover the firmware **212**. In this regard, the processor may determine that a firmware update **218** for the hardware component **210** is stored in the memory **208**. In some examples, the processor may replace the firmware **212** with the firmware update **218** at the hardware component **210**. The processor may authorize the hardware component **210** to establish connection to a bus **216** based on the firmware update **218**.

(47) According to examples, in response to a determination that the firmware update for the hardware component is absent from the memory, the processor may disallow connection of the hardware component **210** to the bus **216**, disable the hardware component **210**, and/or generate a notification indicating that the authentication of the firmware **212** has failed. In some examples, the processor may determine that the authentication of the firmware **212** has failed in response to a determination that the firmware signature **214** does not match the firmware **212** or in response to a determination that the firmware signature **214** is absent from the memory **208**.

(48) For simplicity and illustrative purposes, the present disclosure is described by referring mainly

to examples. In the previous description, numerous specific details are set forth in order to provide a thorough understanding of the present disclosure. It will be readily apparent however, that the present disclosure may be practiced without limitation to these specific details. In other instances, some methods and structures have not been described in detail so as not to unnecessarily obscure the present disclosure.
(49) Throughout the present disclosure, the terms “a” and “an” are intended to denote at least one of a particular element. As used herein, the term “includes” means includes but not limited to, the term “including” means including but not limited to. The term “based on” means based at least in part on.
(50) Although described specifically throughout the entirety of the instant disclosure, representative examples of the present disclosure have utility over a wide range of applications, and the above discussion is not intended and should not be construed to be limiting, but is offered as an illustrative discussion of aspects of the disclosure.
(51) What has been described and illustrated herein is an example of the disclosure along with some of its variations. The terms, descriptions and figures used herein are set forth by way of illustration and are not meant as limitations. Many variations are possible within the scope of the disclosure, which is intended to be defined by the following claims—and their equivalents—in which all terms are meant in their broadest reasonable sense unless otherwise indicated.

## Claims

1. An apparatus comprising: a hardware component; a memory; and a processor to: retrieve a firmware for the hardware component subsequent to the firmware being loaded on the hardware component; determine whether a firmware signature associated with the firmware is stored in the memory; in response to a determination that the firmware signature associated with the firmware is absent from the memory, initiate a recovery process to recover the firmware on the hardware component, wherein to initiate the recovery process, the processor is to: disallow connection of the hardware component to a bus; disable the hardware component; replace the firmware for the hardware component; and generate a notification indicating an authentication status of the firmware of the hardware component; and in response to a determination that the firmware signature associated with the firmware is stored in the memory, initiate authentication of the firmware.
2. The apparatus of claim 1, wherein the processor is to determine that the firmware is not authentic in response to the determination that the firmware signature associated with the firmware is not stored in the memory.
3. The apparatus of claim 1, wherein, in response to a determination that the firmware signature associated with the firmware is stored in the memory, the processor is further to retrieve the firmware signature from the memory to a basic input/output system (BIOS) to authenticate the firmware.
4. The apparatus of claim 1, wherein the processor is to: determine whether a firmware update associated with the hardware component is stored in the memory; and in response to a determination that the firmware is not authentic, initiate a firmware replacement process to replace the firmware with the firmware update.
5. The apparatus of claim 1, wherein the memory is a storage of an embedded controller.
6. The apparatus of claim 5, wherein: the memory and the firmware signature are stored in the embedded controller; and the processor is to execute a BIOS to allow or disallow connection of the hardware component during bootup of the apparatus in response to the firmware being authenticated using the firmware signature.
7. An apparatus comprising: a hardware component; a basic input/output system (BIOS); an embedded controller having a memory; and a processor to execute the BIOS to: retrieve a firmware for the hardware component; determine whether the firmware is inauthentic in response to a

firmware signature associated with the firmware being absent from the memory of the embedded controller; and in response to the determination that the firmware is inauthentic, initiate a recovery process to recover the firmware for the hardware component, wherein to initiate the recovery process, the process is to execute the BIOS to: disallow connection of the hardware component to a bus; disable the hardware component; replace the firmware for the hardware component; and generate a notification indicating an authentication status of the firmware of the hardware component.

8. The apparatus of claim 7, wherein the processor is to execute the BIOS to further determine that the firmware is inauthentic in response to a verification process to authenticate the firmware using the firmware signature associated with the firmware.

9. The apparatus of claim 7, wherein a firmware update associated with the hardware component is stored in the memory, and in response to a determination that the firmware is inauthentic, the processor is to execute the BIOS to recover the firmware on the hardware component by replacing the firmware with the firmware update.

10. A non-transitory computer-readable medium on which is stored computer-readable instructions that when executed cause a processor of a computing device to: retrieve a firmware for a hardware component of the computing device at bootup of the computing device, the firmware being previously updated at the hardware component; initiate authentication of the firmware using a firmware signature stored in a memory; in response to a determination that the authentication of the firmware has failed, initiate a recovery process, wherein to initiate the recovery process the instructions further cause the processor to: disallow connection of the hardware component to a bus; disable the hardware component determine that a firmware update for the hardware component is stored in the memory; replace the firmware with the firmware update at the hardware component; generate a notification indicating an authentication status of the firmware of the hardware component; and authorize the hardware component to establish a connection to the bus based on the firmware update.

11. The non-transitory computer-readable medium of claim 10, wherein the instructions are further to cause the processor to: determine that the authentication of the firmware has failed in response to a determination that the firmware signature does not match the firmware or in response to a determination that the firmware signature is absent from the memory.